Project - High Level Design

on

sports-content-generator

Course Name: generative-AI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr no	Student Name	Enrolment Number
1	ABHAY SINGH THAKUR	EN22CS304003
2	SARTHAK GOKHALE	EN22CS304057
3	HIMANSHI PORWAL	EN22CS306021
4	AKASH MUVEL	EN23EE3L1001
5	PRATHAM JOSHI	EN22CS306037

Group Name: ***Group 04D11***

Project Number: ***GAI-30***

Industry Mentor Name: ***Mr. Rishab Tripathi***

*University Mentor Name:****Dr Hemlata patel***

*Academic Year:****2022-2026***

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used (*If applicable)*

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement & Objectives

1. Problem Statement

In today's digital era, sports content such as match reports, summaries, and articles are generated manually by sports journalists and content writers. This process requires time, domain knowledge, and writing expertise. With the increasing number of sports events happening daily, manual content creation becomes slow, expensive, and difficult to scale. There is a need for an automated system that can generate sports content quickly and efficiently using artificial intelligence.

The main problem addressed in this project is the automation of sports content generation using Generative AI. The system should accept structured inputs such as sport type, match title, highlights, tone, and content length, and automatically generate well-structured sports articles, summaries, reports, or blogs in natural language. The system should also improve the quality of generated content by retrieving relevant context using vector similarity search.

2. Project Objectives

The main objectives of the Sports_GenAI project are:

- To design and develop a web-based application that automatically generates sports content using Generative AI.
- To create a structured input system where users can enter match details such as sport type, match title, highlights, tone, and content length
- To implement a FastAPI backend that processes user input and coordinates different services.
- To implement a prompt builder that converts structured input into a well-formatted prompt suitable for AI content generation.
- To implement a vector retrieval system using FAISS to retrieve relevant sports context.

- To integrate a Large Language Model (Gemini API) to generate natural language sports content.
- To design a modular architecture including frontend, backend, vector service, prompt builder, and LLM service.
- To provide a user-friendly dashboard interface for content generation.
- To demonstrate the use of Generative AI, vector embeddings, and prompt engineering in a real-world application.

3. Scope of the Project

The scope of the Sports_GenAI project includes the development of a web-based system that generates sports content using Generative AI. The system allows users to input match-related data and generate sports articles in different formats and tones. The project includes frontend development, backend development, AI integration, and vector database integration.

The project supports multiple sports such as cricket, football, basketball, and tennis. It supports multiple content types such as articles, summaries, reports, and blogs. The system uses Gemini API for text generation and FAISS vector database for context retrieval. The system is designed as a prototype and focuses mainly on content generation functionality.

The current scope does not include user authentication, database storage for generated articles, mobile application support, or real-time match data integration. These features can be added in future versions of the project.

Proposed Solution

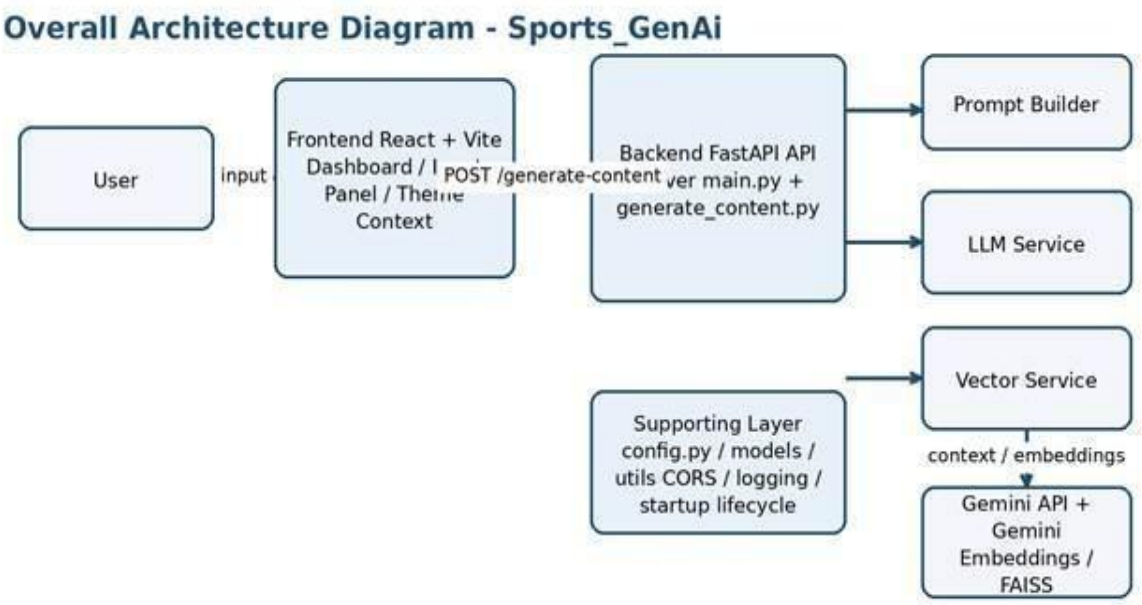
1. Key Features

The key features of the Sports_GenAI system are:

1. Automatic sports content generation using Generative AI.
2. Structured input form for match details.
3. Support for multiple sports.
4. Support for multiple content types such as article, summary, report, and blog.
5. Tone selection for generated content (professional, exciting, analytical, casual).
6. Context retrieval using FAISS vector database.
7. Prompt builder for structured prompt generation.
8. FastAPI backend for handling API requests.
9. React-based frontend dashboard.
10. REST API communication between frontend and backend.
11. Health check API for backend monitoring.
12. Modular layered architecture for scalability and maintainability.

2. Overall Architecture / Workflow

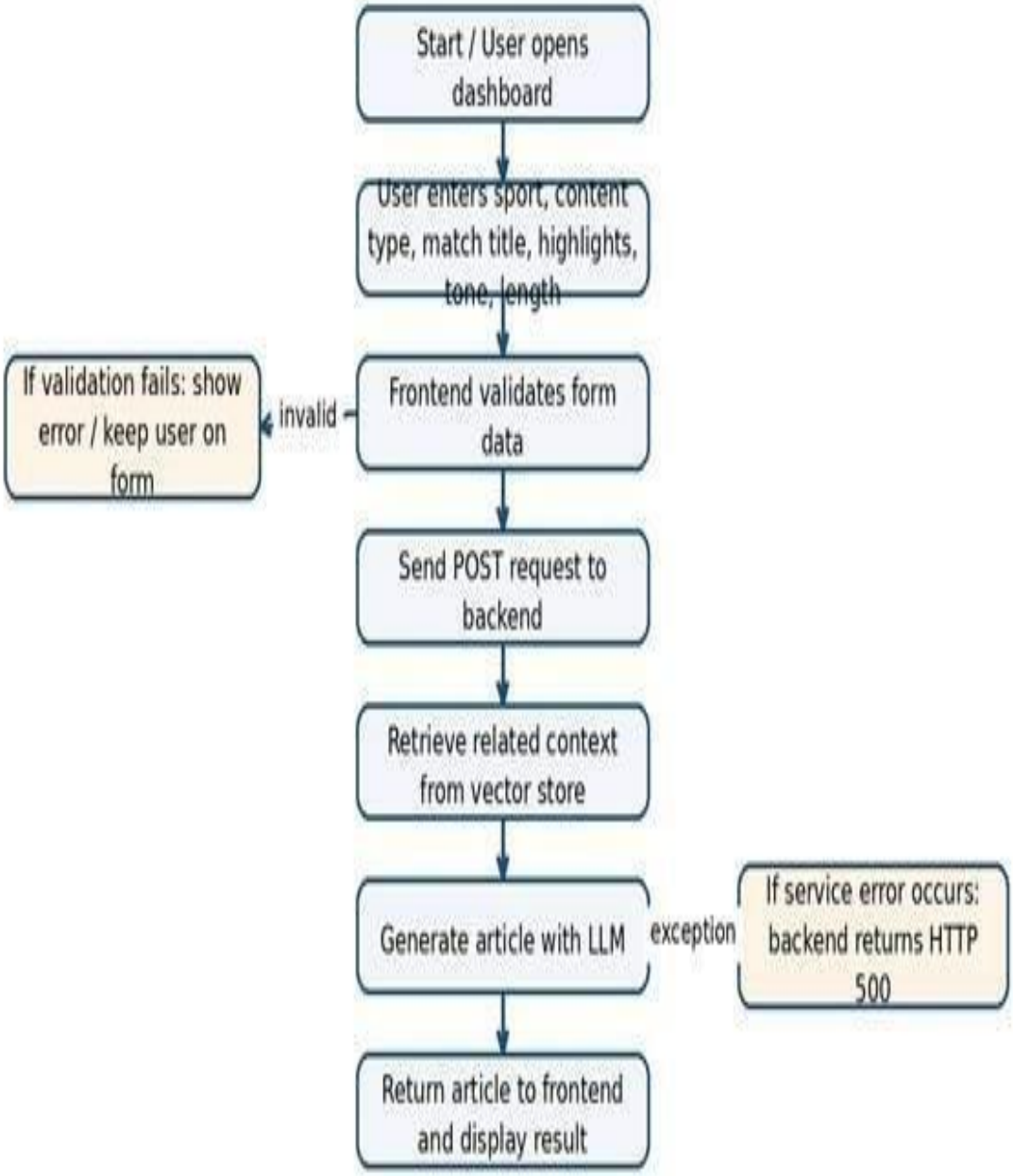
The Sports_GenAI system follows a layered architecture consisting of Presentation Layer, Application Layer, Service Layer, and Support Layer.



Step-by-Step Workflow:

1. The user opens the Sports_GenAI web dashboard.
2. The user enters the required inputs such as sport, content type, match title, highlights, tone, and length.
3. The frontend validates the input fields and sends a POST request to the backend API.
4. The FastAPI backend receives the request and validates the request data using request models.
5. The backend sends the query to the vector service to retrieve relevant sports context.
6. The vector service uses embeddings and FAISS vector database to perform similarity search.
7. The retrieved context is sent to the prompt builder.
8. The prompt builder combines user input and retrieved context and creates a structured sports journalism style prompt.
9. The prompt is sent to the LLM service.
10. The LLM service calls the Gemini API to generate sports content.
11. The generated content is returned to the backend.
12. The backend sends the generated article to the frontend in JSON format.
13. The frontend displays the generated content to the user.

Process Flow / Activity Diagram



4. Tools & Technologies Used

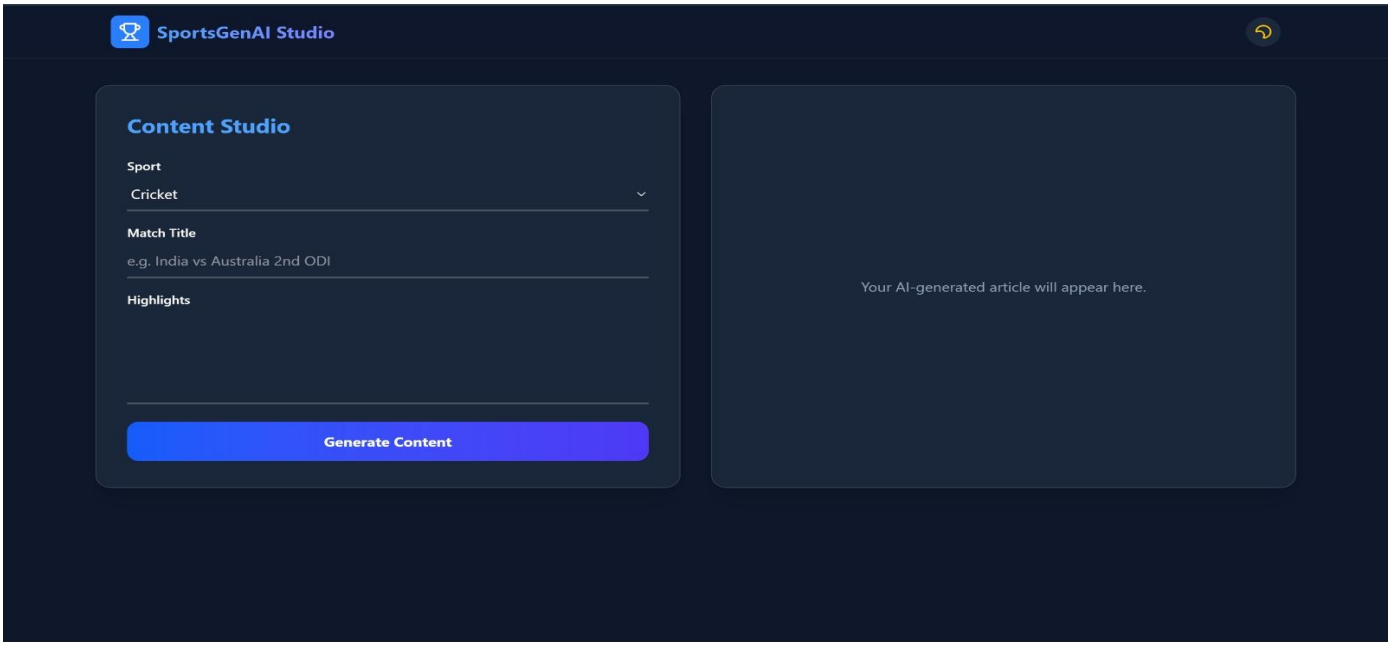
Category	Technology	Purpose
Frontend	React	Used to build the user interface and dashboard for user interaction
Frontend Build Tool	Vite	Used for fast frontend development and project bundling
Styling	Tailwind CSS	Used for designing and styling the user interface
Backend	FastAPI	Used to build REST API and handle backend logic
Programming Language	Python	Used for backend development, AI integration, and API creation
Programming Language	JavaScript	Used for frontend development and API calls
LLM	Google Gemini API	Used to generate sports content using Generative AI
Embeddings	Gemini Embeddings	Used to convert text into vector embeddings for similarity search
Vector Database	FAISS	Used to store embeddings and retrieve relevant context using similarity search
API Communication	REST API	Used for communication between frontend and backend
Environment	Python venv	Used for dependency and environment management
Version Control	Git & GitHub	Used for source code management and version tracking

Results & Output

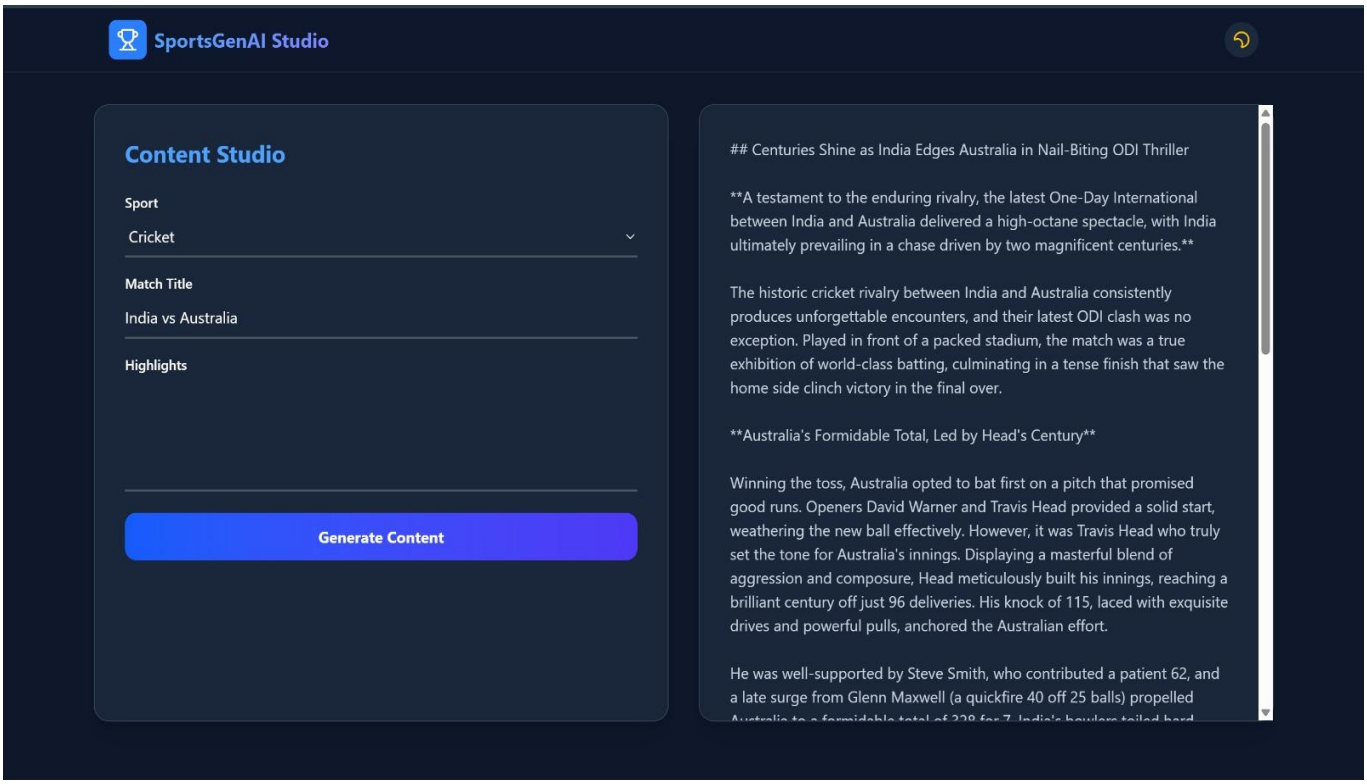
1. Screenshots / Outputs

The project produces the following outputs:

- Dashboard interface where the user enters match details.



- AI-generated sports article output, Generated sports article displayed



2. Reports / Dashboards / Models

The project includes the following system components:

- Web dashboard for content generation.
- Prompt builder module for prompt generation.
- Vector retrieval module using FAISS.
- LLM integration module using Gemini API.
- FastAPI backend service.
- REST API endpoints for content generation.
- Generated sports articles as output.

3. Key Outcomes

The key outcomes of this project are:

- Successfully developed a Generative AI-based sports content generation system.
- Implemented a layered architecture including frontend, backend, and service layer.
- Integrated vector database (FAISS) for context retrieval.
- Integrated Gemini LLM for natural language content generation.
- Implemented prompt engineering for structured sports content generation.
- Developed REST APIs for frontend-backend communication.
- The system can generate sports articles in different tones and formats.
- The project demonstrates real-world use of Generative AI and Retrieval-Augmented Generation (RAG).

Conclusion

The Sports_GenAI project is a web-based generative AI application designed to automatically generate sports content from structured user inputs. The system uses a React-based frontend and FastAPI backend, along with FAISS vector database for context retrieval and Gemini Large Language Model for text generation. The project follows a modular layered architecture which improves maintainability, scalability, and flexibility.

Through this project, we learned about Generative AI, prompt engineering, vector embeddings, retrieval-augmented generation (RAG), FastAPI development, frontend-backend integration, REST API design, and system architecture design. The project demonstrates how AI can be used to automate content creation in the sports domain and reduce manual effort.

Future Scope & Enhancements

The future scope of the Sports_GenAI project includes:

- Adding user authentication and login system.
- Storing generated articles in a database.
- Adding more sports and more content formats.
- Improving prompt engineering for better content quality.
- Deploying the system using Docker and Kubernetes.
- Adding multilingual content generation.
- Integrating real-time sports APIs for live match data.
- Creating a mobile application version.
- Adding analytics dashboard for generated content.
- Using cloud vector databases like Pinecone or Weaviate.
- Adding caching for faster response time.
- Adding user history and saved articles feature.